

Syllabus - CSE 30264 - Computer Networks - Fall 2026

Overview	This course introduces students to fundamental topics on the principles, design, implementation, and performance of computer networks. Topics include: Internet architecture, protocols, socket programming, congestion control, switching and routing, local area networks, mobile and ad-hoc networks, network security, the end-to-end arguments and resource allocation.
Lecture	MW 12:30 - 1:45, Jenkins Nanovic B071
Instructor	Prof. Aaron Striegel, 211B Cushing Hall striegel@nd.edu, 574 631-6896 Office Hours: See Canvas, On Demand (stop by, e-mail, coordinate via Slack)
TAs	Annapurna Puttaswamy, aputtasw@nd.edu
Pre-Requisite	CSE 20289: Systems Programming (or equivalent), Waiver on discussion with instructor
Textbook	James. F. Kurose, Keith W. Ross, "Computer Networking: A top-down approach", 8th or 9th edition (preferred)
Resources	Canvas + Class Repository (Github)

Course Objectives

Upon successful completion of the course, students will be able to:

- Articulate fundamental computer network principles
- Describe the key algorithms and protocols that drive the Internet
- Reason about network setups of varying complexity and modalities
- Conduct basic and advanced implementations of network programming
- Conduct and review network experiments including interpretation and analysis

Grading

Coding Project 1	15%	Mid-Term Exam	25%
Coding Project 2	5%	Final Exam - Written	40%
Assignments	5%	Final Exam - Oral	10%

- Assignments and projects are typically due 11:59 PM (just before midnight), Canvas time.
 - Assignments should be done **individually**, projects may be done in a group of up to **four** students. Group submissions require identification of individual contributions.
- Inquiries about graded materials must be made within one week after return.
- The late policy is as follows:
 - 25% off the maximum score per day (ceiling function), Canvas submission time / late calculation governs. A five-minute grace period is allowed.

- Two exams for the class (midterm - October 12th, final - TBA).
 - One page of notes (front / back side) will be allowed for the mid-term, two front / back pages for the final exam.
 - The oral portion of the final exam consist of a 15 minute session with Prof. Striegel during finals week. Same study sheet as final exam.

Grade range will follow the normal range and will be listed on Canvas.

		94+	A	90-94	A-
87-90	B+	84-87	B	80-84	B-
77-80	C+	74-77	C	70-74	C-
60-70	D	Below 60	F		

In the event of poor class performance on a graded item, the score for that item may be curved. The curve will always result in an improved score for the class, never a reduction.

Course Policies

Inclusive Excellence: As a Catholic research university, Notre Dame is committed to defending the dignity of every human person, to promoting a just society in which every person can flourish, and to attending particularly to the needs of the most vulnerable. This class shares that commitment.

Honor Code: As a member of the Notre Dame community, you acknowledge that it is your responsibility to learn and abide by principles of intellectual honesty and academic integrity, and therefore you will not participate in nor tolerate academic dishonesty.

Usage of AI: AI tools are not allowed for this class for assignments and projects. Inadvertent AI tool usage (ex. Google Gemini via search) should be avoided if possible. If a submission is deemed to be reliant on AI by the instructor or TAs, it will receive a zero. If you are unsure, it is your responsibility to reach out to the instructor.

While AI tools can be helpful, this is not the class to hone your AI skills. The purpose of this class is to teach you networking fundamentals. We will actively examine the limits of AI tools as part of the class for networking fundamentals, protocols, and security.

Many of the course assignments and projects will have intentional vagueness. Such vagueness often plays havoc with LLMs but will make you better engineers and scientists by: (1) Capturing the complex interactions of networked systems / protocols that occur in practice and (2) Encouraging deep thinking and reflection.

Deadlines: Be pro-active to the extent possible. Start projects early and embrace a work-conserving approach to reading and assignments. Extensions are generally granted when requested sufficiently early as the specific situation warrants.

Lecture: Lecture will be recorded upon request with sufficient advanced notice. In-person lecture is highly recommended. Past courses generally show a 5-7% final grade improvement for those who regularly attend lecture.